PES-VCS Lab Report

Name: Aditya Raj
SRN: PES2UG24CS033
Project: Orange Version Control System (PES-VCS)
Date: April 16, 2026

---

1. Build Output

```
Build Output

adityaraj@MacBook-Pro ~/osorange % make clean && make all
gcc -I/opt/homebrew/opt/openssl@3/include -Wall -Wextra -O2 -c object.c -o object.o
gcc -I/opt/homebrew/opt/openssl@3/include -Wall -Wextra -O2 -c tree.c -o tree.o
gcc -I/opt/homebrew/opt/openssl@3/include -Wall -Wextra -O2 -c index.c -o index.o
gcc -I/opt/homebrew/opt/openssl@3/include -Wall -Wextra -O2 -c commit.c -o commit.o
gcc -I/opt/homebrew/opt/openssl@3/include -Wall -Wextra -O2 -c pes.c -o pes.o
gcc -L/opt/homebrew/opt/openssl@3/lib -o pes object.o tree.o index.o commit.o pes.o -lcrypto
gcc -I/opt/homebrew/opt/openssl@3/include -Wall -Wextra -O2 -c test_objects.c -o test_objects.o
gcc -L/opt/homebrew/opt/openssl@3/lib -o test_objects test_objects.o object.o -lcrypto
gcc -I/opt/homebrew/opt/openssl@3/include -Wall -Wextra -O2 -c test_tree.c -o test_tree.o
gcc -L/opt/homebrew/opt/openssl@3/lib -o test_tree test_tree.o object.o tree.o -lcrypto
```

---

2. Phase 1: Object Storage

2A. Test Output

```
Phase 1: ./test_objects

adityaraj@MacBook-Pro ~/osorange % ./test_objects
Stored blob with hash: d58213f5dbe0629b5c2fa28e5c7d4213ea09227ed0221bbe9db5e5c4b9aafc12
Object stored at: .pes/objects/d5/8213f5dbe0629b5c2fa28e5c7d4213ea09227ed0221bbe9db5e5c4b9aafc12
PASS: blob storage
PASS: deduplication
PASS: integrity check

All Phase 1 tests passed.
```

2B. Object Store Structure

```
Phase 1: object store

adityaraj@MacBook-Pro ~/osorange % find .pes/objects -type f | sort
.pes/objects/25/ef1fa07ea68a52f800dc80756ee6b7ae34b337afedb9b46a1af8e11ec4f476
.pes/objects/2a/594d39232787fba8eb7287418aec99c8fc2ecdaf5aaf2e650eda471e566fcf
.pes/objects/d5/8213f5dbe0629b5c2fa28e5c7d4213ea09227ed0221bbe9db5e5c4b9aafc12
```

---

3. Phase 2: Tree Objects

3A. Test Output

```
Phase 2: ./test_tree

adityaraj@MacBook-Pro ~/osorange % ./test_tree
Serialized tree: 139 bytes
PASS: tree serialize/parse roundtrip
PASS: tree deterministic serialization

All Phase 2 tests passed.
```

3B. Tree Object Hex Dump

```
Phase 2: tree hex dump

adityaraj@MacBook-Pro ~/osorange % xxd .pes/objects/f5/902ee73b6aa25436f687067fc4cc0cb481cf416740d3c3c3ed6d9c4de7b6c2 | head -20
00000000: 7472 6565 2039 3800 3130 3036 3434 2066  tree 98.100644 f
00000010: 696c 6531 2e74 7874 002c f8d8 3d9e e295  ile1.txt.,..=...
00000020: 43b3 4a87 7274 21fd ecb7 e3f3 a183 d337  C.J.rt!........7
00000030: 6390 25de 576d b9eb b431 3030 3634 3420  c.%.Wm...100644 
00000040: 6669 6c65 322e 7478 7400 e00c 50e1 6a2d  file2.txt...P.j-
00000050: f38f 8d6b f809 e181 ad02 48da 6e67 19f3  ...k......H.ng..
00000060: 5f9f 7e65 d6f6 0619 9f7f                 _.~e......
```

---

4. Phase 3: Index (Staging Area)

4A. Commands Output

```
Phase 3: pes status

adityaraj@MacBook-Pro ~/osorange % ./pes init
adityaraj@MacBook-Pro ~/osorange % ./pes add file1.txt file2.txt
adityaraj@MacBook-Pro ~/osorange % ./pes status
Initialized empty PES repository in .pes/
Staged changes:
  staged:     file1.txt
  staged:     file2.txt

Unstaged changes:
  (nothing to show)

Untracked files:
  untracked:  index.h
  untracked:  object.c
  untracked:  pes.c
  untracked:  index_demo_b.txt
  untracked:  report.md
  untracked:  Makefile
  untracked:  index_demo_a.txt
  untracked:  test_objects
  untracked:  tree.h
  untracked:  test_objects.c
  untracked:  commit.c
  untracked:  test_tree
  untracked:  README.md
  untracked:  index.c
  untracked:  test_sequence.sh
  untracked:  .gitignore
  untracked:  render_terminal_image.py
  untracked:  pes_asan
  untracked:  test_tree.c
  untracked:  pes.h
  untracked:  report.pdf
  untracked:  commit.h
  untracked:  tree.c
```

4B. Index File Content

```
Phase 3: .pes/index

adityaraj@MacBook-Pro ~/osorange % cat .pes/index
100644 2cf8d83d9ee29543b34a87727421fdecb7e3f3a183d337639025de576db9ebb4 1776334706 6 file1.txt
100644 e00c50e16a2df38f8d6bf809e181ad0248da6e6719f35f9f7e65d6f606199f7f 1776334706 6 file2.txt
```

---

5. Phase 4: Commits

5A. Commit Log

```
Phase 4: pes log

adityaraj@MacBook-Pro ~/osorange % export PES_AUTHOR="Aditya Raj <PES2UG24CS033>"
adityaraj@MacBook-Pro ~/osorange % ./pes commit -m "Initial snapshot"
adityaraj@MacBook-Pro ~/osorange % ./pes log
Committed: 78e289248dba... Initial snapshot
commit 78e289248dba11a3452cfccee6210278f44c41d07bb4247e9f53a5084cf1790f
Author: Aditya Raj <PES2UG24CS033>
Date:   1776334706

    Initial snapshot
```

5B. .pes Directory Structure

```
Phase 4: .pes files

adityaraj@MacBook-Pro ~/osorange % find .pes -type f | sort
.pes/HEAD
.pes/index
.pes/objects/2c/f8d83d9ee29543b34a87727421fdecb7e3f3a183d337639025de576db9ebb4
.pes/objects/78/e289248dba11a3452cfccee6210278f44c41d07bb4247e9f53a5084cf1790f
.pes/objects/e0/0c50e16a2df38f8d6bf809e181ad0248da6e6719f35f9f7e65d6f606199f7f
.pes/objects/f5/902ee73b6aa25436f687067fc4cc0cb481cf416740d3c3c3ed6d9c4de7b6c2
.pes/refs/heads/main
```

5C. HEAD and References

```
Phase 4: HEAD refs

adityaraj@MacBook-Pro ~/osorange % cat .pes/HEAD
adityaraj@MacBook-Pro ~/osorange % cat .pes/refs/heads/main
ref: refs/heads/main

78e289248dba11a3452cfccee6210278f44c41d07bb4247e9f53a5084cf1790f
```

---

6. Final Integration Test

```
Final: make test-integration

adityaraj@MacBook-Pro ~/osorange % make test-integration
=== Running integration tests ===
bash test_sequence.sh
=== PES-VCS Integration Test ===

--- Repository Initialization ---
Initialized empty PES repository in .pes/
PASS: .pes/objects exists
PASS: .pes/refs/heads exists
PASS: .pes/HEAD exists

--- Staging Files ---
Status after add:
Staged changes:
  staged:     file.txt
  staged:     hello.txt

Unstaged changes:
  (nothing to show)

Untracked files:
  (nothing to show)


--- First Commit ---
Committed: f58cf4fee2d7... Initial commit

Log after first commit:
commit f58cf4fee2d7812a8ab4acd0fb308f7587e7f1e946b07d71a823550734eadef2
Author: Aditya Raj <PES2UG24CS033>
Date:   1776334706

    Initial commit


--- Second Commit ---
Committed: a1b28c27f19c... Update file.txt

--- Third Commit ---
Committed: 280bf13a0efd... Add farewell

--- Full History ---
commit 280bf13a0efd0b50726a0d3281314461fd264ce01ae910b8ca9568db89d8e30c
Author: Aditya Raj <PES2UG24CS033>
Date:   1776334707

    Add farewell

commit a1b28c27f19ce6e26970b40b8f9e64adabb241c2a6b9e46c7550c728c0657463
Author: Aditya Raj <PES2UG24CS033>
Date:   1776334706

    Update file.txt

commit f58cf4fee2d7812a8ab4acd0fb308f7587e7f1e946b07d71a823550734eadef2
Author: Aditya Raj <PES2UG24CS033>
Date:   1776334706

    Initial commit


--- Reference Chain ---
HEAD:
ref: refs/heads/main
refs/heads/main:
280bf13a0efd0b50726a0d3281314461fd264ce01ae910b8ca9568db89d8e30c

--- Object Store ---
Objects created:
      10
.pes/objects/0b/d69098bd9b9cc5934a610ab65da429b525361147faa7b5b922919e9a23143d
.pes/objects/10/1f28a12274233d119fd3c8d7c7216054ddb5605f3bae21c6fb6ee3c4c7cbfa
.pes/objects/28/0bf13a0efd0b50726a0d3281314461fd264ce01ae910b8ca9568db89d8e30c
.pes/objects/58/a67ed1c161a4e89a110968310fe31e39920ef68d4c7c7e0d6695797533f50d
.pes/objects/a1/b28c27f19ce6e26970b40b8f9e64adabb241c2a6b9e46c7550c728c0657463
.pes/objects/ab/5824a9ec1ef505b5480b3e37cd50d9c80be55012cabe0ca572dbf959788299
.pes/objects/b1/7b838c5951aa88c09635c5895ef7e08f7fa1974d901ce282f30e08de0ccd92
.pes/objects/d0/7733c25d2d137b7574be8c5542b562bf48bafeaa3829f61f75b8d10d5350f9
.pes/objects/db/07a1451ca9544dbf66d769b505377d765efae7adc6b97b75cc9d2b3b3da6ff
.pes/objects/f5/8cf4fee2d7812a8ab4acd0fb308f7587e7f1e946b07d71a823550734eadef2

=== All integration tests completed ===
```

7. Analysis Questions

Q5.1 Branch Checkout

A branch is stored as a reference file pointing to a commit hash. Implementing checkout would involve:

- Updating `HEAD` to point to the new branch
- Reconstructing the working directory from the target commit tree
- Overwriting files carefully

The complexity arises in handling uncommitted changes.

Q5.2 Dirty Working Directory

Compare:

- Working directory files
- Index entries
- Target commit tree

If differences exist in tracked files, block checkout to avoid data loss.

Q5.3 Detached HEAD

`HEAD` points directly to a commit instead of a branch.
New commits are created but not referenced by any branch.
Recovery is possible by creating a new branch pointing to that commit.

Q6.1 Garbage Collection

- Traverse all reachable commits from refs
- Mark reachable objects using DFS/BFS
- Delete unmarked objects

A hash set can be used to track visited objects.

Q6.2 GC Race Condition

If GC runs during commit:

- It may delete objects that are not yet referenced

Git avoids this using locking and reference safety.